

Malhar Gudekar

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MASTER ACHIEVEMENTS BANK

University of Illinois Urbana-Champaign — Research Assistant (iSchool)

- Owned end-to-end design, development, and integration of cross-platform system ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized PostgreSQL schema, indexing strategies, and query execution plans for high-volume datasets, improving query performance by 38% under production workloads
- Built robust, production-ready Python pipeline with retrieval and context summarization, improving multi-turn response accuracy by 43%
- Developed backend workflows for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

CHI AI in Infodemic Management — Research Assistant

- Designed and deployed highly scalable NLP pipelines in Python to process high-volume datasets, reducing processing latency by 41%
- Built distributed data ingestion pipelines using Kafka and Neo4j Streams for real-time, high-throughput processing and pattern detection
- Designed and developed production-grade REST APIs using FastAPI, implementing validation, monitoring, and deployment workflows for scalable integration
- Enabled large-scale data processing across social media streams using distributed system architecture principles

Business Intelligence Group — Technical Consultant

- Built and deployed scalable microservices-based Retrieval-Augmented Generation (RAG) system using FastAPI, PostgreSQL, and vector embeddings for production applications
- Designed efficient, fault-tolerant ETL pipelines to process and index large document datasets for low-latency retrieval systems
- Improved system performance by 64% through optimized indexing strategies and pipeline efficiency improvements in production environments
- Developed backend services enabling real-time, high-performance querying of unstructured data across distributed system architectures

Swift Mobil Software Solutions — Data Analyst

- Built interactive Power BI dashboards for logistics and mobility KPIs, reducing reporting time by 40% and improving visibility into operational performance
- Processed and optimized large-scale datasets using Python and PySpark to identify failure patterns, trends, and performance issues in distributed environments
- Developed SQL and graph queries to map dependencies, analyze operations, detect data quality issues, accelerating root-cause analysis by 26%

Pscope Technologies — Data Analyst

- Developed and maintained 20+ Power BI dashboards across Sales, Finance, and Operations using SQL and DAX, enabling KPI tracking and data-driven decision-making
- Automated recurring Excel-based reporting workflows using VBA, implementing validation and scheduling logic, reducing report preparation time by 42%

Stu/Dio at Illinois — Project Manager

- Led cross-functional team of 6 engineers and designers to deliver product updates using user feedback, telemetry data, and performance insights for feature prioritization
- Tracked sprint progress, delivery metrics, and QA issues through JIRA-based workflows, managing backlog grooming and ensuring on-time delivery

Scalable ML System for Customer Risk Prediction

- Built end-to-end machine learning system using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale time-series data
- Designed scalable ETL and feature engineering pipelines capturing behavioral trends and temporal patterns to improve model performance
- Developed decision framework for credit line adjustments, improving prediction precision by 20% and enabling risk-aware classification

FMCG Sales Demand Forecasting and Optimization

- Developed ARIMA-based sales forecasting model in R achieving 80% accuracy with 95% predictions within confidence intervals across five FMCG categories
- Used ggplot2 and dplyr to analyze daily sales trends, improving inventory planning and reducing stockouts and overstock scenarios

FMCG Product Sales Analysis

- Engineered Tableau dashboard analyzing 12+ product categories across 100+ countries, identifying key markets with revenue peaks of \$200M
- Built multi-view performance tracking system identifying 30% sales volatility and enabling early detection of major dips and recoveries
- Developed geographic heat maps and filters revealing underperforming categories and supporting targeted improvement strategies

Advanced Facial Analysis System

- Built real-time facial recognition system using Python, OpenCV, and CNN achieving 98% emotion classification accuracy
- Processed 10,000+ images to train supervised learning model achieving 95% accuracy across varying lighting and orientations

Basilisk Cryptography System

- Developed terminal-based encrypted communication system integrating multiple cryptographic algorithms for secure key exchange
- Designed layered encryption mechanism combining up to six algorithms under unified key architecture
- Published research paper titled BASILISK Security Hub in UGC Care Journal Volume 13

Environmental Data Analytics and Visualization

- Analyzed environmental datasets to uncover 20% rise in acetaldehyde levels over seven years using trend visualization techniques
- Built heat maps and tree maps across six states identifying high toxin concentrations and regional risk patterns

Database Design for Nonresident Alien Payments

- Designed relational database with 15+ entities and implemented EER model with SQL and XML schema validation for IRS compliance
- Built Neo4j graph database with automated Cypher query generation using Python and SQLAlchemy for cross-system insights

Mobile Expense Tracker Application

- Developed Flutter application with Provider state management to track and visualize personal expenses
- Integrated dynamic charts and responsive UI enabling weekly spending analysis and improved user experience

AI-Powered AR Virtual Try-On (Glossier)

- Conducted customer segmentation and behavioral analysis identifying high engagement among users aged 18–44 for AR adoption
- Developed strategic roadmap outlining AR integration, ROI projections, and KPIs including return reduction and conversion improvement

Predictive Credit Limit Adjustment

- Built LightGBM-based model forecasting customer spending and enabling personalized credit line adjustments
- Segmented customers into risk groups improving fraud detection and financial decision-making accuracy

F1 Race Prediction System

- Engineered dataset from FastF1 API (2022–2025) to build lap-level and race-level predictive models
- Applied GBM and Monte Carlo simulations to predict race outcomes achieving 90% accuracy for final grid positions